Phylogenetic Relationships Among Five Penguin Species Inferred from COX1 Protein Sequences Using Bayesian, Maximum Likelihood, and Parsimony Methods

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Abstract

Penguins (family Spheniscidae) are a diverse group of flightless, aquatic birds whose evolutionary relationships remain a broad focus in phylogenetics. This study examined genetic diversity and phylogenetic relationships among five penguin species — *Aptenodytes forsteri, Aptenodytes patagonicus, Eudyptula minor, Pygoscelis adeliae*, and *Pygoscelis papua* — using the mitochondrial cytochrome c oxidase subunit I (COX1) gene. Protein sequences were retrieved from the NCBI Gene database and aligned using MAFFT. Phylogenetic trees were constructed using five methods: Bayesian inference (MrBayes), maximum likelihood (IQ-TREE), coalescent-based species tree estimation (ASTRAL), maximum parsimony, and distance-based analysis, with final visualization in iTOL and RStudio. All five methods recovered a mostly consistent topology, placing *Eudyptula minor* as the outgroup and resolving *Aptenodytes forsteri* and *A. patagonicus* as a strongly supported clade, sister to *Pygoscelis adeliae* and *P. papua*. The distance-based NJ tree had *P. papua* as a sister to *P. adeliae*; however, this is likely due to human error rather than a biological signal. These results suggest that COX1 provides a robust phylogenetic signal for resolving relationships among these penguin genera, despite the limitation of using a single gene.

Introduction

Penguins are one of the very few birds that are flightless and aquatic. These adaptations in these birds are seen as evolutionary traits, allowing the species to thrive and survive in harsh conditions (Pelegrin, 2022). This study focuses on five distinct penguin species — *Aptenodytes forsteri, Aptenodytes patagonicus, Eudyptula minor, Pygoscelis adeliae, and Pygoscelis papua* — to examine their genetic diversity, biological mechanisms, and evolutionary significance.

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The five species of penguins selected are part of the Spheniscidae family (Australian Antarctic Program, 2024). *A. forsteri* (Emperor Penguin) and *A. patagonicus* (King Penguin) are by far the largest species of penguins, known for their diving ability (National Geographic, n.d.) While *Eudyptula minor* (Little Penguin) is the smallest species of penguins, hence the name “Little Penguin.” This species was used as a critical evolutionary comparison point. The other two species, *P. adeliae* (Adélie Penguin) and *P. papua* (Gentoo Penguin), are important to understand the evolutionary divergence within brush-tailed penguins, being a genus that occupies a distinct ecological niche from the larger *Aptenodytes* species.

Understanding the phylogenetic relationships among these five species is significant because it reveals morphological and ecological differences, such as body size, diving depth, and habitat preference, that map onto evolutionary history. Resolving these relationships with molecular data provides reasonable framework than just morphology alone, which can be misleading due to convergent evolution. To understand the relationship between these taxa, the mitochondrial gene used for this study was the cytochrome c oxidase subunit I (COX1), as all penguin species have it (Heller, 2018). The COX1 gene is widely used as a starting point in phylogenetic analysis for penguins, staying almost identical for millions of years. Yet, this genetic marker has variation that scientists can use to differentiate penguin species from one another. In this study, the aligned COX1 sequences had no gaps, providing a clean dataset for phylogenetic analysis.

It is hypothesized that the five penguins will exhibit consistent patterns of genetic divergence in maximum-likelihood (ML) phylogenetic trees constructed with Iterative Quick Tree (IQ-TREE), Bayesian inference of phylogeny (MrBayes), maximum parsimony, and distance-based NJ with branch lengths reflecting the degree of evolutionary divergence among species. Coalescent-based analysis using the Accurate Species TRee ALgorithm (ASTRAL) will be employed to further confirm species relationships and account for incomplete lineage sorting (ILS).

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Methods

Data Collection

Protein sequences (COX1) for five different penguin species (*A. forsteri, A. patagonicus, Eudyptula minor, P. adeliae,* and *P. papua*) were collected from the NCBI Gene database (NCBI, 2024a-e). This database was chosen for its reliability and standardized sequence formatting. Sequences were downloaded in FASTA format and manually compiled into a single multi-sequence FASTA file for downstream alignment.

MAFFT

Sequences were aligned using Multiple Alignment using Fast Fourier Transform (MAFFT; Katoh et al., 2002). MAFFT uses Fast Fourier Transform to rapidly find homologous regions between sequences, making it significantly faster than traditional progressive alignment methods, supporting high accuracy for protein sequences. MAFFT was chosen for its speed, accuracy, and wide adoption in phylogenetic studies of mitochondrial protein sequences. The software assumes that input sequences are orthologous and share global homology. A limitation is that MAFFT may produce suboptimal alignments for highly divergent sequences where gap penalties do not perfectly reflect biological reality. Default parameters were used, producing a cleaned alignment of 516 amino acid positions with no gaps across all five taxa.

Distance and Parsimony Trees

Maximum parsimony and distance-based neighbor-joining (NJ) trees were constructed in RStudio using the `phangorn` package (Schliep, 2011). Maximum parsimony identifies the phylogenetic tree that requires the fewest total evolutionary changes, assuming all character state transitions have equal weight. The NJ method calculates pairwise genetic distances between sequences and clusters taxa accordingly. Both methods were included to provide complementary, computationally efficient perspectives on phylogenetic relationships and to assess topological robustness across approaches. Trees were rooted at *Eudyptula minor* as the outgroup using the `root () ` function in the `ape` package.

MrBayes

Bayesian phylogenetic inference was performed using MrBayes 3.2.7 (Huelsenbeck & Ronquist, 2001; Ronquist & Huelsenbeck, 2003). MrBayes uses Markov Chain Monte Carlo (MCMC) sampling to approximate the posterior probability distribution of phylogenetic trees, estimating the likelihood of different tree topologies given the data and substitution model. This method was chosen because posterior probability values provide an intuitive and statistically rigorous measure of branch support, and because Bayesian inference performs well with protein sequence data. MrBayes assumes that the MCMC chains have reached convergence and that the substitution model adequately describes the evolutionary process. A key limitation is that MrBayes is computationally expensive, and insufficient run length can produce misleading posterior probabilities.

The following parameters were used: amino acid model prior set to mixed (aamodelpr=mixed), allowing the software to sample across substitution models; 1,000,000 MCMC generations (ngen=1000000); sampling every 500 generations (samplefreq=500); four parallel chains (nchains=4); and a burnin of 250 samples (burnin=250).

IQ-TREE

Maximum likelihood (ML) phylogenetic inference was performed using IQ-TREE2 (Nguyen et al., 2015; Minh et al., 2020). IQ-TREE uses a fast stochastic algorithm to search for the tree topology that maximizes the likelihood of seeing the sequence data under a given substitution model. IQ-TREE was chosen for its speed, accuracy, and built-in model selection. ModelFinder Plus (-m MFP) was used to automatically find the best-fit substitution model before tree inference. Branch reliability was assessed using 1,000 ultrafast bootstrap replicates. IQ-TREE assumes that the selected substitution model accurately captures the biological evolution of the sequences. A limitation is that ML inference can be sensitive to the starting tree and computationally intensive for larger datasets.

ASTRAL

Species tree estimation was performed using ASTRAL (Mirarab, 2019), which takes the gene tree produced by IQ-TREE as input and estimates the species tree under the multispecies coalescent model. ASTRAL was chosen because it accounts for incomplete lineage sorting, a known source of conflict between gene trees and species trees. However, a major limitation of this study is the use of a single gene (COX1); ASTRAL is designed for multi-gene datasets, and here it primarily functions to visualize uncertainty in the single-gene phylogenetic signal. ASTRAL assumes that gene tree discordance is caused by incomplete lineage sorting rather than other processes such as horizontal gene transfer or hybridization. Default parameters were used.

Tree Visualization

MrBayes, ASTRAL, and IQ-TREE's phylogenetic trees were generated using iTOL (Interactive Tree Of Life; Letunic & Bork, 2021) for final visualization and formatting. iTOL allows consistent scaling of branch lengths and the clear display of support values (bootstrap, posterior probability, and local posterior probability) at the nodes. Maximum parsimony and NJ-Distance trees were generated in RStudio, using the package `apes`.

Results

The Maximum Parsimony tree (Figure 4) produced a topology that broadly aligns with the other methods, naming *Eudyptula minor* as the outgroup. This method effectively grouped the *Aptenodytes* and *Pygoscelis* genera, though the specific branching order within the *Pygoscelis* group can vary based on character weighting. The alignment of 516 amino acid positions provided 0 gaps, ensuring that parsimony-informative sites were consistently available across all five taxa.

However, a minor topological inconsistency was seen in the distance-based (NJ) tree (Figure 5), where *P. papua* was placed outside the *P. adeliae* and *Aptenodytes* clade, rather than as a sister taxon to *P. adeliae* as recovered by all other methods. All other relationships, including the placement of *Eudyptula minor* as the outgroup and the monophyly of the *Aptenodytes* clade, remained consistent across methods (Anthropic, 2025).

Yet, as seen in the Bayesian and ML trees (Figure 1-2), *A. forsteri* and *A. patagonicus* exhibited the shortest genetic distances to one another, confirming their close evolutionary relationship. *Eudyptula minor* stayed the most genetically distant lineage, positioned at the base of the tree. The consistency across all four methods (Bayesian, ML, Coalescent, and Parsimony) strongly suggests that the single COX1 gene is a robust marker for resolving the relationships among these specific penguin genera.

Species	Gene	Aligned Length (bp)	GC Content (%)	Gaps
Aptenodytes forsteri	COX1	516	9.3	0
Aptenodytes patagonicus	COX1	516	9.1	0
Eudyptula minor	COX1	516	9.3	0
Pygoscelis adeliae	COX1	516	9.3	0
Pygoscelis papua	COX1	516	9.3	0

Table 1. Summary of COX1 mitochondrial sequences for five penguin species aligned using MAFFT. All sequences are 516 bp in length with no gaps, showing a clean and complete alignment.

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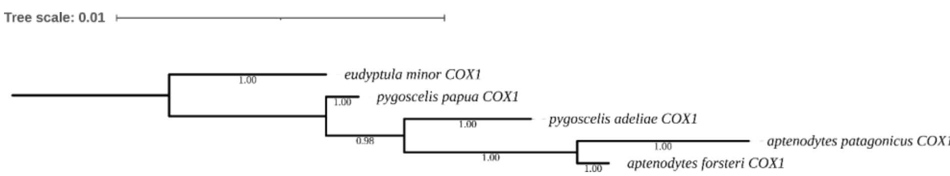

Figure 1. Bayesian inference phylogenetic tree of five penguin species inferred from COX1 mitochondrial sequences using MrBayes. Posterior probability values are shown at each node (values range from 0.98 to 1.00), indicating strong support for all relationships. Branch lengths are proportional to expected substitutions per site (scale bar = 0.01).

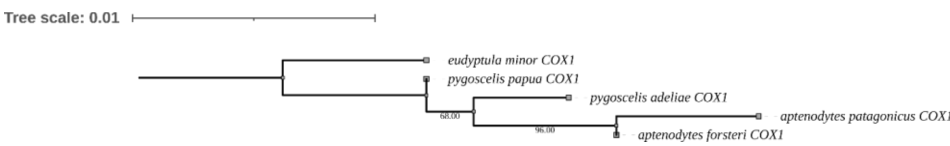

Figure 2. Maximum-likelihood phylogenetic tree of five penguin species, inferred from COX1 mitochondrial sequences using IQ-TREE with ModelFinder Plus (-m MFP) and 1000 ultrafast bootstrap replicates. Bootstrap support values are shown at each node (68.00 and 96.00). Branch lengths are proportional to the number of substitutions per site (scale bar = 0.01).

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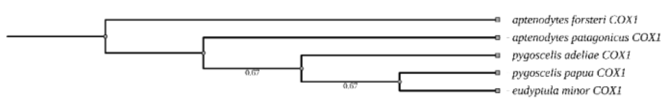

Figure 3. Species tree of five penguin species estimated using ASTRAL under the multispecies coalescent model, inferred from COX1 mitochondrial gene trees generated by IQ-TREE. Local posterior probability values are shown at each node (0.67 at two nodes), showing moderate support for the inferred relationships. Branch lengths are coalescent units.

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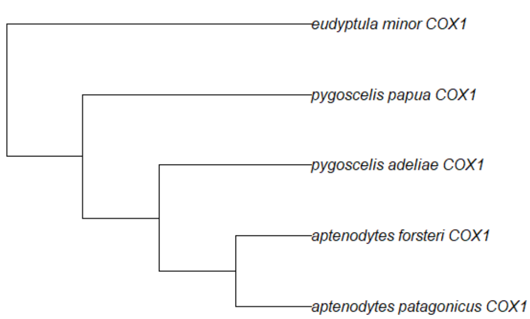

Figure 4. Maximum parsimony phylogenetic tree of five penguin species inferred from COX1 protein sequences in R. *Eudyptula minor* is placed as the outgroup. *Aptenodytes forsteri* and *A. patagonicus* form a well-supported clade, sister to *Pygoscelis adeliae* and *P. papua*, consistent with Bayesian and ML topologies.

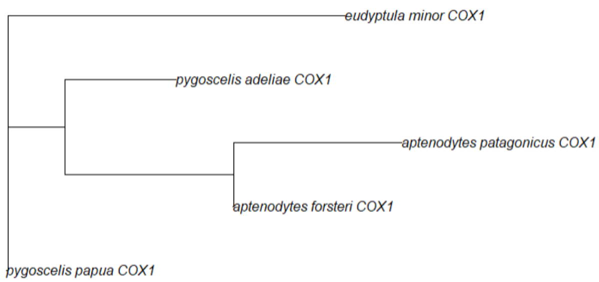

Figure 5. Distance-based neighbor-joining (NJ) phylogenetic tree of five penguin species inferred from COX1 protein sequences in R. Branch lengths are pairwise genetic distances. While *Eudyptula minor* is recovered as the outgroup and the *Aptenodytes* clade is intact, *Pygoscelis papua* is placed outside its expected sister relationship with *Pygoscelis adeliae*, being a minor topological incongruence compared to other methods.

Discussion

This study aimed to analyze the relationship among the five different penguin species using the COX1 gene. The methods used observed a consistent topology (Figure 1-4). This suggests the COX1 gene provides a strong phylogenetic signal for distinguishing the various species used. MAFFT produced a clean matrix of 516 amino acid positions with no gaps (Table 1). GC content is consistently low across all taxa, which is typical for mitochondrial COX1 sequences. This shows strong sequence conservation across the five species used for this study. Likewise, *Eudyptula minor* was the outgroup, as it was the most evolutionary distinct from the rest.

The close relationship between *A. forsteri* and *A. patagonicus* was supported by the high bootstrap (Figure 2, 96.00) and posterior probabilities (Figure 1, 1.00), showing their shared ancestry within the same genus. ML tree (Figure 2) showed a bootstrap value of 96.00 and 68.00, suggesting strong support for the genus-level groupings. ASTRAL's tree (Figure 3) had posterior probability values of 0.67 at two nodes, showing the inherent uncertainty of using a single gene in a coalescent framework.

However, a minor discrepancy was seen in the distance-based NJ tree, where *Pygoscelis papua* was not recovered as sister to *P. adeliae*, likely reflecting a known limitation of distance-based methods. Unlike Bayesian, ML, and parsimony approaches, NJ relies on pairwise genetic distances rather than character-based analysis, making it more sensitive to variation in a small dataset of only five taxa. The consistent placement of *P. papua* as sister to *P. adeliae* across the four remaining methods suggests this discrepancy is an error rather than a true biological signal.

Future studies suggest adding more genes, such as cytochrome c oxidase subunit 2 (COX2) or the NADH dehydrogenase subunit 2 (ND2). This allows ASTRAL to show significant improvements, as it typically runs multiple genes together to make multiple gene trees. Additionally, running Bayesian Evolutionary Analysis Sampling Trees (BEAST) shows divergence dating. BEAST is vital for future studies because it provides a statistically rigorous framework that simultaneously integrates genetic, temporal, and spatial data to model complex evolutionary processes while accurately accounting for uncertainty in massive genomic datasets.

This study demonstrates that COX1 protein sequences provide a reliable phylogenetic signal for determining evolutionary relationships among penguin species. The consistent recovery of *Eudyptula minor* as the outgroup across all five methods aligns with its classification as the most morphologically distinct penguin genus, being the smallest species and the only one native to temperate regions (Heller, 2018). The strong support for *Aptenodytes* monophyly is consistent with earlier molecular studies of penguin systematics (Du et al., 2019), reinforcing the evolutionary cohesion of the two largest penguin species. The moderate ASTRAL support values and the NJ inconsistency both highlight the inherent limitations of single-gene analysis, underscoring the need for multi-gene datasets in future work. Overall, this study confirms the utility of COX1 as a barcode marker for penguin phylogenetics while finding clear directions for methodological improvement.

References

Anthropic. (2025). Claude (claude-sonnet-4-6) [Large language model]. https://claude.ai

Australian Antarctic Program. (2024, May 24). Emperor penguin. Www.antarctica.gov.au. https://www.antarctica.gov.au/about-antarctica/animals/penguins/emperor-penguin/

Bui Quang Minh, Heiko A Schmidt, Olga Chernomor, Dominik Schrempf, Michael D Woodhams, Arndt von Haeseler, Robert Lanfear, IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era, *Molecular Biology and Evolution*, Volume 37, Issue 5, May 2020, Pages 1530–1534, https://doi.org/10.1093/molbev/msaa015

Du, J., Tian, J. S., Lu, Z. C., Zhang, S. J., Song, X. R., Liu, G. Y., & Han, J. B. (2019). Identification of the complete mitochondrial genome of the king penguin Aptenodytes patagonicus (Sphenisciformes: Spheniscidae: Aptenodytes). Mitochondrial DNA Part B, 4(2), 2191-2192.

Fredrik Ronquist, John P. Huelsenbeck, MrBayes 3: Bayesian phylogenetic inference under mixed models, *Bioinformatics*, Volume 19, Issue 12, August 2003, Pages 1572–1574, https://doi.org/10.1093/bioinformatics/btg180

John P. Huelsenbeck, Fredrik Ronquist, MRBAYES: Bayesian inference of phylogenetic trees Bioinformatics, Volume 17, Issue 8, August 2001, Pages 754–755, https://doi.org/10.1093/bioinformatics/17.8.754

Kazutaka Katoh, Kazuharu Misawa, Kei-ichi Kuma, Takashi Miyata, MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform, *Nucleic Acids Research*, Volume 30, Issue 14, 15 July 2002, Pages 3059–3066, https://doi.org/10.1093/nar/gkf436

Lam-Tung Nguyen, Heiko A. Schmidt, Arndt von Haeseler, Bui Quang Minh, IQ-TREE: A Fast and Effective Stochastic Algorithm for Estimating Maximum-Likelihood Phylogenies, *Molecular Biology and Evolution*, Volume 32, Issue 1, January 2015, Pages 268–274, https://doi.org/10.1093/molbev/msu300

Letunic, I., & Bork, P. (2021). Interactive Tree Of Life (iTOL) v5: An online tool for phylogenetic tree display and annotation. *Nucleic Acids Research*, *49*(W1), W293–W296. https://doi.org/10.1093/nar/gkab301

Mirarab, S. (2019). Species Tree Estimation Using ASTRAL: Practical Considerations.

arXiv:1904.03826

Heller, P., Casaletto, J., Ruiz, G., & Geller, J. (2018). A database of metazoan cytochrome c oxidase subunit I gene sequences derived from GenBank with CO-ARBitrator. *Scientific data*, *5*, 180156. https://doi.org/10.1038/sdata.2018.156

National Center for Biotechnology Information (NCBI). (2024). *Cytochrome c oxidase subunit I (COX1), Aptenodytes forsteri* [Gene ID: 26045193]. NCBI Gene. https://www.ncbi.nlm.nih.gov/datasets/gene/26045193

National Center for Biotechnology Information (NCBI). (2024). *Cytochrome c oxidase subunit I (COX1), Aptenodytes patagonicus* [Gene ID: 26737703]. NCBI Gene. https://www.ncbi.nlm.nih.gov/datasets/gene/26737703

National Center for Biotechnology Information (NCBI). (2024). *Cytochrome c oxidase subunit I (COX1), Eudyptula minor* [Gene ID: 806254]. NCBI Gene. https://www.ncbi.nlm.nih.gov/datasets/gene/806254

National Center for Biotechnology Information (NCBI). (2024). *Cytochrome c oxidase subunit I (COX1), Pygoscelis adeliae* [Gene ID: 36935527]. NCBI Gene. https://www.ncbi.nlm.nih.gov/datasets/gene/36935527

National Center for Biotechnology Information (NCBI). (2024). *Cytochrome c oxidase subunit I (COX1), Pygoscelis papua* [Gene ID: 42904018]. NCBI Gene. https://www.ncbi.nlm.nih.gov/datasets/gene/42904018

National Geographic. (n.d.). *Emperor penguin*. https://www.nationalgeographic.com/animals/birds/facts/emperor-penguin.

Pelegrín, J. S., & Acosta Hospitaleche, C. (2022). Evolutionary and biogeographical history of penguins (Sphenisciformes): Review of the dispersal patterns and adaptations in a geologic and paleoecological context. Diversity, 14(4), 255.

Schliep, K. P. (2011). phangorn: Phylogenetic analysis in R. *Bioinformatics*, *27*(4), 592–593. https://doi.org/10.1093/bioinformatics/btq706